



Analytics Solutions Bootcamp: #05

# Data Science and Machine Learning Essentials

Examine how models are built through experimentation and feature design.

Hello, everyone! Welcome to the fifth module of the Analytics Solutions Bootcamp. Today, you will learn about the essentials of data science and machine learning.

# Topics to be covered



- What is data science and machine learning?
- Data science workflow
- EDA and data preparation for machine learning
- Predictive analytics
  - Supervised machine learning
  - Unsupervised machine learning
  - Model evaluation, overfitting, underfitting
- Prescriptive analytics

# What is data science and machine learning?

Understand what data science is and how we use machine learning for building analytics solutions

September 11, 2026

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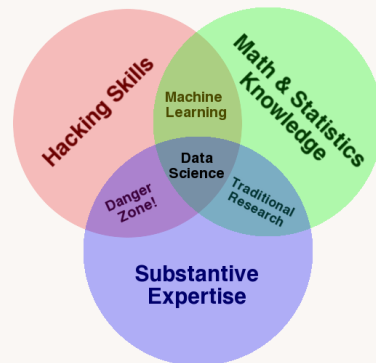
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What is data science and machine learning?

## Data science defined



Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data.

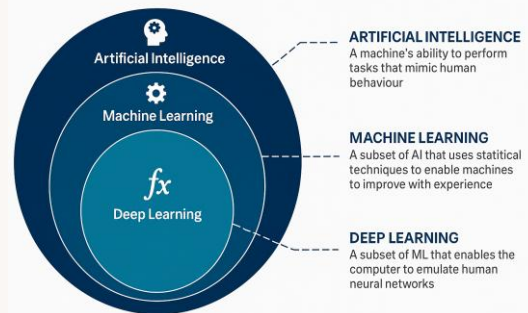


What is data science and machine learning?

## Machine learning defined



Machine learning is the subset of artificial intelligence (AI) focused on algorithms that can “learn” the patterns of training data and, subsequently, make accurate inferences about new data.



# Data science workflow

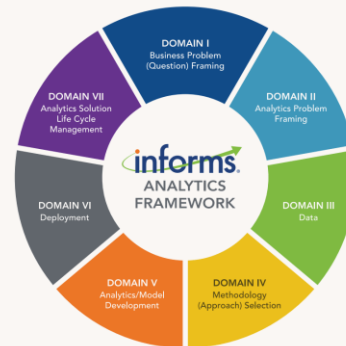
Learn how to perform data science projects

# The data science workflow



The INFORMS analytics framework shows the steps of doing data science projects.

We will use this framework throughout this module.



# I. Business problem framing



This includes activities related to understanding the business problem and relevant questions, as well as evaluating the scope of the problem.

Ex. "What are the common characteristics of the survivors from Titanic? Would I survive?"

```

> titanic_data <- read.csv("https://raw.githubusercontent.com/datasciencedojo/datasets/refs/heads/master/titanic.csv")
> str(titanic_data)
'data.frame':   891 obs. of  12 variables:
 $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Survived   : int  0 1 1 1 0 0 0 1 1 ...
 $ Pclass     : int  3 1 3 1 3 3 3 2 ...
 $ Name       : chr  "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"
 $ Sex        : chr  "male" "female" "female" "female" ...
 $ Age        : num  22 38 26 35 35 NA 54 2 27 14 ...
 $ SibSp      : int  1 1 0 1 0 0 0 1 0 1 ...
 $ Parch      : int  0 0 0 0 0 0 1 2 0 ...
 $ Ticket     : chr  "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
 $ Fare       : num  7.25 71.28 7.92 53.1 8.05 ...
 $ Cabin      : chr  "" "C85" "" "C123" ...
 $ Embarked   : chr  "S" "C" "S" ...

> head(titanic_data)
  PassengerId Survived Pclass                    Name
1          1         0       3 Braund, Mr. Owen Harris
2          2         1       1 Cumings, Mrs. John Bradley (Florence Briggs Thayer)
3          3         1       3 Heikkinen, Miss. Laina
4          4         1       1 Futrelle, Mrs. Jacques Heath (Lily May Peel)
5          5         0       3 Allen, Mr. William Henry
6          6         0       3 Moran, Mr. James

  Sex Age SibSp Parch Ticket   Fare Cabin Embarked
1 male  22     1     0   A/5 21171  7.2500    S
2 female 38     1     0   PC 17599 71.2833    C
3 female 26     0     0 STON/O2. 3101282 7.9250    S
4 female 35     1     0 113803 53.1000 C123    S
5 male  35     0     0 373450  8.0500    S
6 male  NA     0     0 330877  8.4583    Q

```

## II. Analytics problem framing



Translates a clearly defined business problem (question) into a structured analytics problem that can be addressed with analytics techniques applied to data.

Ex. "What characteristics were common among survivors? Based on my characteristics, what would have been my estimated probability of survival?"

```

> titanic_data <- read.csv("https://raw.githubusercontent.com/datasciencedojo/datasets/refs/heads/master/titanic.csv")
> str(titanic_data)
'data.frame':   891 obs. of  12 variables:
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 $ Survived   : int  0 1 1 0 0 0 0 1 1 ...
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 $ Sex        : chr  "male" "female" "female" "female" ...
 $ Age       : num  22 38 26 35 35 NA 54 2 27 14 ...
 $ SibSp     : int  1 1 0 1 0 0 0 1 0 1 ...
 $ Parch     : int  0 0 0 0 0 0 1 2 0 ...
 $ Ticket    : chr  "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
 $ Fare      : num  7.25 71.28 7.92 53.1 8.05 ...
 $ Cabin     : chr  "" "C85" "" "C123" ...
 $ Embarked  : chr  "S" "C" "S" ...

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5           5         0       3 Allen, Mr. William Henry
6           6         0       3 Moran, Mr. James

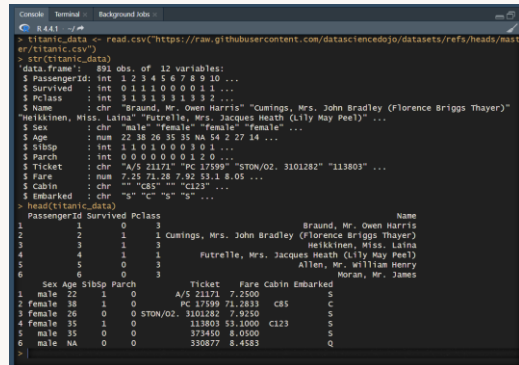
  Sex Age SibSp Parch Ticket   Fare Cabin Embarked
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4 female 35     1     0 113803 53.1000 C123    S
5 male  35     0     0 373450  8.0500    S
6 male  NA     0     0 330877  8.4583    Q

```

### III. Data: Identifying data sources >>

This step encompasses activities related to identifying the data required and available, ensuring its usability for solving analytics problems; and fulfilling documentation and reporting needs.

Ex. Titanic dataset



```

> titanic_data <- read.csv("https://raw.githubusercontent.com/datasciencedojo/datasets/refs/heads/master/titanic.csv")
> str(titanic_data)
'data.frame':   891 obs. of  12 variables:
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 $ Parch      : int  0 0 0 0 0 0 1 2 0 ...
 $ Ticket     : chr  "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
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```

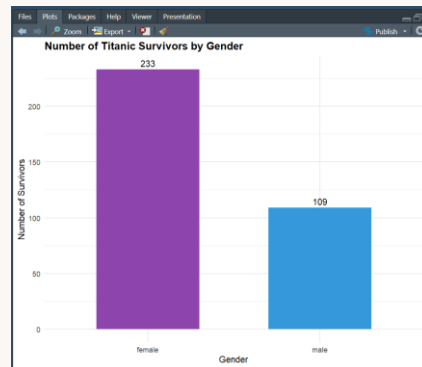
Mention data management plan

Data science workflow

### III. Data: Exploratory data analysis >>

Exploratory data analysis helps in investigating each independent variable and how it relates with the dependent variable.

Ex. Survivor count by sex



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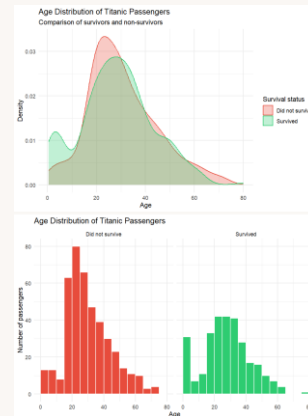
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Mention data management plan

## III. Data: Exploratory data analysis >>

Exploratory data analysis helps in investigating each independent variable and how it relates with the dependent variable.

Ex. Distribution of age of survivors vs non-survivors



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Mention data management plan

## III. Data: Split training and test >>

Split training and test datasets. 80% on training is arbitrary and can be adjusted. Specific sampling techniques may be needed. Set the seed when dealing with randomization.

```
24 # 3. Split train and test set
25 set.seed(123)
26
27
28 titanic_split <- initial_split(
29   titanic_model,
30   prop = 0.80,
31   strata = Survived
32 )
33
34 train_data <- training(titanic_split)
35 test_data <- testing(titanic_split)
```

Mention data management plan

### III. Data: Feature engineering



Perform imputations on missing data, create dummy variables as needed, and perform other feature engineering techniques that may be needed prior to model training.

```
# Recipe for feature engineering
titanic_recipe <- recipe(
  Survived ~ .,
  data = train_data
)

# Impute missing values with median
step_impute_median(all_numeric_predictors()) |>

# Impute missing categories with most frequent mode
step_impute_mode(all_nominal_predictors()) |>

# Handle unexpected categories in future data
step_novel(all_nominal_predictors()) |>

# Convert sequential variables into dummy variables
step_dummy(all_nominal_predictors()) |>

# Remove variables with no variation
step_rm(all_predictors())

prepared_recipe <- prep(
  titanic_recipe,
  training = train_data
)

train_processed <- bake(
```

```
#> glimpse(train_processed)
Rows: 712
Columns: 12
$ Age      <dbl> 35, 29, 54, 2, 20, 39, 14, 31, 35, 8, 29, 19,...
$ SibSp    <dbl> 0, 0, 0, 3, 0, 1, 0, 1, 0, 3, 0, 1, 1,...
$ Parch    <dbl> 0, 0, 0, 1, 0, 3, 0, 0, 0, 1, 0, 2, 0,...
$ FamilySize <dbl> 1, 1, 1, 5, 1, 7, 1, 2, 1, 5, 1, 6, 1, 2, 2,...
$ Fare     <dbl> 8.0500, 8.4583, 51.8625, 21.0750, 8.0500, 31,...
$ Survived <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ Pclass_X2 <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,...
$ Pclass_X3 <dbl> 1, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0,...
$ Sex_male <dbl> 1, 1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1,...
$ isAlone_yes <dbl> 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0,...
$ Embarked_S <dbl> 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ Embarked_5 <dbl> 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 0, 0, 1,...
```

Mention data management plan

## IV. Methodology selection



This involves selecting the appropriate analytical methods, tools, and technologies to solve the defined analytics problem.

Ex. Logistic regression for binary classification

```
Console Terminal Background Ids
> titanic_data <- read.csv("https://raw.githubusercontent.com/datasciencedojo/datasets/refs/heads/main/titanic.csv")
> str(titanic_data)
'data.frame':   891 obs. of  12 variables:
 $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
 $ Survived   : int  0 1 1 1 0 0 0 1 1 ...
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 $ Sex        : chr  "male" "female" "female" "female" ...
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> head(titanic_data)
  PassengerId Survived Pclass                    Name
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5 male  35     0     0 373450 8.0500   S
6 male  NA     0     0 330877 8.4583   Q
```

## V. Analytics / model development>>

This covers activities related to designing, building, and refining models, as well as documenting and communicating the results.

Ex. Result of the logistic regression model

```
> titanic_logistic <- glm(
+   Survived ~ .,
+   data = train_processed,
+   family = binomial(link = "logit")
+ )
> summary(titanic_logistic)

Call:
glm(formula = Survived ~ ., family = binomial(link = "logit"),
    data = train_processed)

Coefficients: (1 not defined because of singularities)
              (Intercept)  4.511134  0.570962  7.901 2.77e-15 ***
              Age        -0.038231  0.008957  -4.268 1.97e-05 ***
              SibSp       -0.487284  0.174642  -2.790  0.00527 **
              Parch       -0.301867  0.150471  -2.006  0.04484 *
              FamilySize   NA         NA         NA
              Fare        0.001470  0.002471  0.095  0.53183
              Pclass_X2   -0.980178  0.324441  -3.021  0.00252 **
              Pclass_X3   -2.119397  0.321425  -6.594 4.29e-11 ***
              Sex_male    -2.534664  0.225408  -11.245 < 2e-16 ***
              IsAlone_Yes -0.833895  0.314125  -2.655  0.00794 **
              Embarked_Q  -0.109417  0.434081  -0.252  0.80099
              Embarked_S  -0.233037  0.267557  -0.846  0.34425
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 947.98  on 711 degrees of freedom
Residual deviance: 633.86  on 701 degrees of freedom
AIC: 655.86
```

## V. Analytics / model development>>

After model training, validation can be done. Validation / test set helps in checking if the model performs well.

```
> # 8. Basic evaluation
> # Confusion matrix
> table(
+   Actual = test_predictions$Survived,
+   Predicted = test_predictions$PredictedSurvival
+ )
      Predicted
Actual 0  1
0 95 15
1 16 53
>
> # Accuracy
> mean(
+   test_predictions$Survived ==
+   test_predictions$PredictedSurvival
+ )
[1] 0.8268156
```

```
> test_predictions <- test_processed |>
+   mutate(
+     SurvivalProbability = predict(
+       titanic_logistic,
+       newdata = test_processed,
+       type = "response"
+     ),
+     PredictedSurvival = if_else(
+       SurvivalProbability >= 0.50,
+       1,
+       0
+     )
+   )
>
> test_predictions |>
+   select(
+     Survived,
+     SurvivalProbability,
+     PredictedSurvival
+   ) |>
+   head()
# A tibble: 6 x 3
  Survived SurvivalProbability PredictedSurvival
  <dbl>         <dbl>         <dbl>
1     0         0.153             0
2     1         0.936             1
3     1         0.925             1
4     1         0.928             1
5     0         0.0733            0
6     0         0.0888            0
>
```

## V. Analytics / model development>>

Random spot check can also be done.

Ex. "Male, 35, I'm with my wife and her child. I can only pay the average fare."

```
> prediction
# A tibble: 1 x 2
  SurvivalProbability PredictedOutcome
      <dbl> <chr>
1      0.208 Unlikely to survive
>
```

```
sample_passenger <- tibble(
  Pclass = factor("2", levels = levels(train_data$Pclass)),
  Sex = factor("male", levels = levels(train_data$Sex)),
  Age = 35,
  SibSp = 1,
  Parch = 1,
  FamilySize = 3,
  IsAlone = factor("No", levels = levels(train_data$IsAlone)),
  Fare = mean(train_data$Fare, na.rm = TRUE),
  Embarked = factor("S", levels = levels(train_data$Embarked))
)

sample_processed <- bake(
  prepared_recipe,
  new_data = sample_passenger
)

survival_probability <- predict(
  ultra_logistic,
  newdata = sample_processed,
  type = "response"
)

prediction <- tibble(
  SurvivalProbability = as.numeric(survival_probability),
  PredictedOutcome = if_else(
    survival_probability >= 0.50,
    "likely to survive",
    "unlikely to survive"
  )
)

prediction
```

## V. Deployment



This covers activities related to delivering the analytics solution and ensuring it is effectively integrated into the business environment.

Ex. ML workflow

```
> titanic_logistic <- glm(
+   Survived ~ .,
+   data = train_processed,
+   family = binomial(link = "logit")
+ )
> summary(titanic_logistic)

Call:
glm(formula = Survived ~ ., family = binomial(link = "logit"),
    data = train_processed)

Coefficients: (1 not defined because of singularities)
              (Intercept)  4.511134  0.570962  7.901 2.77e-15 ***
Age                -0.038231  0.008957  -4.268 1.97e-05 ***
 SibSp             -0.487284  0.174642  -2.790  0.00527 **
 Parch             -0.301867  0.150471  -2.006  0.04484 *
 FamilySize        NA         NA         NA         NA
 Fare              0.001470  0.002471  0.095  0.53183
 Pclass_X2         -0.980178  0.324441  -3.021  0.00252 ***
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 Sex_male          -2.534664  0.225408  -11.245 < 2e-16 ***
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 Embarked_Q        -0.109417  0.434081  -0.252  0.80099
 Embarked_S        -0.233037  0.267557  -0.846  0.34425
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 947.98  on 711 degrees of freedom
Residual deviance: 633.86  on 701 degrees of freedom
AIC: 655.86
```

## VI. AS Lifecycle management



This involves ongoing oversight and calibration to ensure the analytics solution continues to perform effectively and delivers valid results over time that continue to deliver business value.

Ex. How often do we rerun the analysis?

```
> titanic_logistic <- glm(
+   Survived ~ .,
+   data = train_processed,
+   family = binomial(link = "logit")
+ )
> summary(titanic_logistic)

Call:
glm(formula = Survived ~ ., family = binomial(link = "logit"),
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Coefficients: (1 not defined because of singularities)
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# EDA and data preparation for machine learning

Inspecting and preparing your data prior to predictive analytics

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Data preparation and EDA for machine learning

## EDA objectives for ML models >>

- Understand the data better
- Uncover insights that show the relationships between independent variables and dependent variables
- Identify possible redundancy on the data
- Check if the data meets model assumptions

## Some terminologies first



- “Variables”: synonymous to “columns”
  - X, input, independent variables, predictors, features
  - Y, output, dependent variables, outcome, response

## Understand and profile data



- Create a data dictionary of the data.
- Perform data profiling for quality checks.
- Identify what variables are relevant to the problem.
- Identify the independent and dependent variables

```
R Console - Background Info
> head(titanic, 20)
# A tibble: 20 x 12
  PassengerId Survived Pclass Name                Sex    Age Sibsp Parch Ticket   Fare Cabin Embarked
  <dbl>      <dbl> <dbl> <chr>              <fchr> <dbl> <dbl> <dbl> <chr>    <dbl> <chr> <fchr>
1         1         0       3 Braund, Mr. Owen Harris male    22   1   0 A/5 2.  7.25 NA   S
2         2         1       1 Cumings, Mrs. John Bradley female  38   1   0 PC 17. 71.3 C85   C
3         3         1       3 Heikkinen, Miss. Laina female  26   0   0 S309./ 53.1 NA   S
4         4         1       1 Futrelle, Mrs. Jacques He. female  35   1   0 113803 53.1 C123  S
5         5         0       3 Allen, Mr. William Henry male    35   0   0 373450 53.05 NA   S
6         6         0       3 Moran, Mr. James male     NA   0   0 330877 8.46 NA   Q
7         7         0       1 McCarthy, Mr. Timothy J male    54   0   0 17463 51.9 E46   S
8         8         0       3 Palsson, Master. Gosta Le. male     2   3   1 349900 21.1 NA   S
9         9         1       1 Johnson, Mrs. Oscar W (El. female  27   0   2 347742 11.1 NA   S
10        10         1       2 Nassar, Mrs. Nicholas (Ad. female  14   1   0 237736 30.1 NA   C
11        11         1       3 Sandstrom, Miss. Marguerite female  4   1   1 PP 95. 16.7 GG   S
12        12         1       1 Bonnell, Miss. Elizabeth female  58   0   0 113783 26.6 C103  S
13        13         0       3 Gaumnard, Mr. William male     20   0   0 A/5.  4.03 NA   S
14        14         0       3 Andersson, Mr. Anders Joh. male    39   1   1 347082 31.3 NA   S
15        15         0       3 Vestrom, Miss. Hilda Amen. female  14   0   0 350406 7.85 NA   S
16        16         1       2 Hewlett, Mrs. Mary D Kin. female  55   0   0 248706 16   NA   S
17        17         0       1 Rice, Master. Eugene male     2   4   1 382652 29.1 NA   Q
18        18         1       2 Williams, Mr. Charles Esp. male    NA   0   0 248323 11   NA   S
19        19         0       3 Vander Planke, Mrs. Julia female  31   1   0 345763 18   NA   S
20        20         1       3 Hasselmann, Mrs. Fritza female  NA   0   0 2949  7.22 NA   C
```

## Dealing with missing data



- Understand the definition of “missing” data (null, N/A, 1900-01-01)
- Decide what you do with missing data
  - Remove column, row, or impute

```
R441 ~/R
> data.frame(
+   missing_rate = sapply(titanic, function(x) mean(is.na(x)))
+ )
      missing_rate
PassengerId 0.000000000
Survived     0.000000000
Pclass       0.000000000
Name         0.000000000
Sex          0.000000000
Age          0.198653199
SibSp        0.000000000
Parch        0.000000000
Ticket       0.000000000
Fare         0.000000000
Cabin        0.771043771
Embarked     0.002244669
>
>
```

## Data imputation techniques



- Simple imputation techniques
  - Mean, median, mode imputation
  - Fixed value / constant imputation
- Advanced imputation techniques
  - K-nearest neighbors
  - Regression imputation
  - Multivariate imputation by chained equations (MICE)

# Perform univariate analysis



## Numerical variables

- Mean and median
- Minimum and maximum
- Distribution and outliers

## Categorical variables

- Frequency and proportions
- Class imbalance

```
> summary(titanic)
  PassengerId  Survived  Pclass     Name     Sex
Min.   :   1.0   Min.   :0.0000   Min.   :1.000   Length:891   Length:891
1st Qu.:223.5   1st Qu.:0.0000   1st Qu.:2.000   Class :character   Class :character
Median :446.0   Median :0.0000   Median :3.000   Mode  :character   Mode  :character
Mean   :446.0   Mean   :0.3838   Mean   :2.359
3rd Qu.:668.5   3rd Qu.:1.0000   3rd Qu.:3.000
Max.   :891.0   Max.   :1.0000   Max.   :3.000

  Age      SibSp     Parch    Ticket     Fare
Min.   : 0.42   Min.   :0.000   Min.   :0.0000   Length:891   Min.   : 0.00
1st Qu.:20.12   1st Qu.:0.000   1st Qu.:0.0000   Class :character   1st Qu.: 7.25
Median :28.00   Median :0.000   Median :0.0000   Mode  :character   Median :14.45
Mean   :29.70   Mean   :0.523   Mean   :0.3836   Mean   :32.20
3rd Qu.:38.00   3rd Qu.:1.000   3rd Qu.:0.0000   3rd Qu.:31.00
Max.   :80.00   Max.   :8.000   Max.   :6.0000   Max.   :512.33
NA's   :177

  Cabin     Embarked
Length:891   Length:891
Class :character   Class :character
Mode  :character   Mode  :character

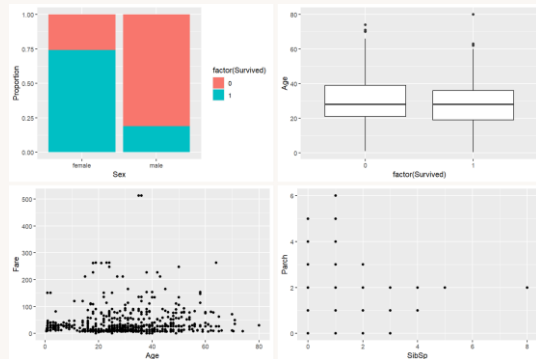
> table(titanic$Survived)
 0  1
549 342
> sd(titanic$Age, na.rm = TRUE)
[1] 14.5265
>
```

# Perform bivariate analysis



## Categorical vs numerical

- Use of bar / columns graphs
- Box and whiskers



## Numerical vs numerical

- Scatterplot
- Multicollinearity

Data preparation and EDA for machine learning

## Feature engineering



Feature engineering is the process of selecting, creating or modifying features like input variables or data to help machine learning models learn patterns more effectively. It involves transforming raw data into meaningful inputs that improve model accuracy and performance.

## Feature creation



Feature creation: Feature creation involves generating new features from domain knowledge or by observing patterns in the data.

```
> titanic %>% select(PassengerId, Name, SibSp, Parch) %>% mutate(FamilySize = SibSp + Parch + 1)
# A tibble: 891 x 5
  PassengerId Name      SibSp Parch FamilySize
  <dbl> <chr> <dbl> <dbl> <dbl>
1      1 Braund, Mr. Owen Harris     1     0       2
2      2 Cumings, Mrs. John Bradley (Florence Briggs Th...   1     0       2
3      3 Heikkinen, Miss. Laina           0     0       1
4      4 Futrelle, Mrs. Jacques Heath (Lily May Peel)       1     0       2
5      5 Allen, Mr. William Henry         0     0       1
6      6 Moran, Mr. James              0     0       1
7      7 McCarthy, Mr. Timothy J         0     0       1
8      8 Palsson, Master. Gosta Leonard       3     1       5
9      9 Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Ber...   0     2       3
10     10 Nasser, Mrs. Nicholas (Adele Achem)       1     0       2
# > 881 more rows
# Use `print(n = ...)` to see more rows
>
```

## Feature transformation



Feature transformation: this involves normalization and scaling of data, encoding categorical variables, and applying mathematical transformations (log transforms).

## Standardization and normalization>>

Standardization: standardizes the data to have a mean of 0 and a standard deviation of 1.

$$z = \frac{x_i - \mu}{\sigma}$$

Normalization: scaling the variable to a new scale. Common scaling technique is min-max scaling.

$$x_{scaled} = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

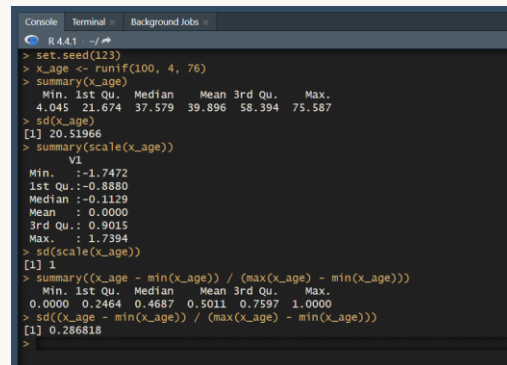
## Standardization and normalization>>

Standardization: standardizes the data to have a mean of 0 and a standard deviation of 1.

$$z = \frac{x_i - \mu}{\sigma}$$

Normalization: scaling the variable to a new scale. Common scaling technique is min-max scaling.

$$x_{scaled} = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$



```
R 4.4.1 -- /R/
> set.seed(123)
> x_age <- runif(100, 4, 76)
> summary(x_age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  4.045  21.674   37.579  39.896  58.394  75.587
> sd(x_age)
[1] 20.51966
> summary(scale(x_age))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -1.7472 -0.8880 -0.1129  0.0000  0.9015  1.7394
> sd(scale(x_age))
[1] 1
> summary((x_age - min(x_age)) / (max(x_age) - min(x_age)))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.2464  0.4687  0.5011  0.7597  1.0000
> sd((x_age - min(x_age)) / (max(x_age) - min(x_age)))
[1] 0.286818
>
```

## Encoding variables



Converts categorical variable to numerical form.

- Label encoding. Converts each text value to a whole number, like 0, 1, 2, 3. Ex. Red = 0, green = 1, yellow = 2
- One-hot encoding. Makes new column for each group.
  - Ex. High school graduate, college graduate, post-grad degree
    - Create binary columns education\_hs, education\_college, education\_postgrad

# Encoding variables example



Label encoding: most applicable for ordinal data

Ordinal encoding: most applicable for nominal data

```
R441 -> R
> titanic_copy <- titanic
> unique(titanic_copy$Pclass)
[1] 3 1 2
> titanic_copy %>% mutate(
+   Embarked_S = if_else(Embarked=="S",1,0),
+   Embarked_C = if_else(Embarked=="C",1,0),
+   Embarked_Q = if_else(Embarked=="Q",1,0),
+   Pclass_rev = 3 + 1 - Pclass
+ ) %>% select(PassengerId, Name, Embarked_S, Embarked_C, Embarked_Q, Pclass, Pclass_rev)
# A tibble: 891 x 7
  PassengerId Name      Embarked_S Embarked_C Embarked_Q Pclass Pclass_rev
  <dbl> <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
1 1 Braund, Mr. Owen Harris 0 0 0 3 1
2 2 Cumings, Mrs. John Bradley (Florence Briggs Thayer) 0 1 0 1 3
3 3 Heikkinen, Miss. Laina 0 0 0 3 1
4 4 Fatoules, Mrs. Jacques Heath (Lily May Peel) 1 0 0 1 3
5 5 Allen, Mr. William Henry 1 0 0 3 1
6 6 Moran, Mr. James 0 0 1 3 1
7 7 McCarthy, Mr. Timothy J 1 0 0 1 3
8 8 Palsson, Master. Gosta Leonard 1 0 0 0 1
9 9 Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg) 1 0 0 1 1
10 10 Nasser, Mrs. Nicholas (Adele Achem) 0 1 0 2 2
# Use print(n = ...) to see more rows
#
```

# Predictive analytics (supervised & unsupervised)

Use machine learning models for predictive analytics and exploratory analysis

Predictive analytics

## Predictive analytics



In predictive analytics, we use the historical data to find patterns and “predict” the future based on these.

The two types of predictive analytics are supervised (labeled) and unsupervised learning (unlabeled).

## Some considerations



- Model: refers to machine learning model used for supervised or unsupervised learning
  - Business requirements and task type
  - Evaluation metrics
  - Explainability vs accuracy
  - Model assumptions
- *Building a machine learning model is easy. Building it correctly is hard.*

# Supervised learning

Use machine learning models analyzing labeled data

Supervised learning

# Supervised learning



In supervised learning, we predict the occurrence of an attribute from other variables.

Given:

$$Y = X$$

Our goal is to estimate for  $f$ :

$$Y = f(X) + \epsilon$$

## Types of supervised learning >>

- Classification: it's a classification problem if our goal is to predict a categorical response variable from one or more predictors. Classification models can be binary or multiclass / multilabel.
- Regression: it's a regression problem if our goal is to predict a numerical response variable from one or more predictors.
  - Forecasting is a specialized extension of regression that predicts future continuous values by analyzing data points sequentially ordered over time.

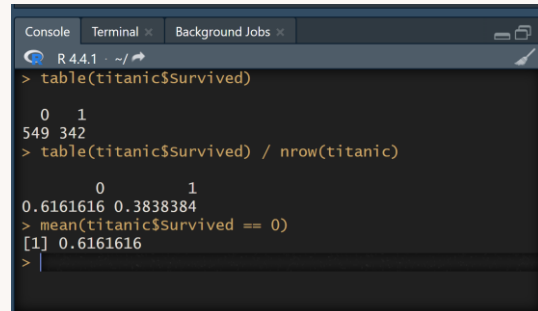
# Classification problem

Learn how we analyze data, and build machine learning models for predicting discrete categorical response

## How do we estimate classes? >>

### Frequency-based classification: Zero rule

- Titanic dataset had 342 (38%) survivors, and 549 (62%) fatalities.
- We predict the majority class.
- This yields 62% prediction accuracy.



```
Console Terminal Background Jobs x
R 4.4.1 ~ /
> table(titanic$Survived)
 0  1
549 342
> table(titanic$Survived) / nrow(titanic)
      0      1
0.6161616 0.3838384
> mean(titanic$Survived == 0)
[1] 0.6161616
> |
```

## How do we estimate classes? >>

### Frequency-based classification: One rule

- Generate one rule for each predictor in the data, and the predictor with the highest accuracy will be the one rule.
- One-rule: female passengers will survive (79% accuracy)

```
Console Terminal Background Jobs
R 4.4.1 - /#
> titanic %>% group_by(Sex, Survived) %>% summarise()
# A tibble: 4 x 3
# Groups:   Sex [2]
  Sex    Survived n()
<chr>    <dbl> <int>
1 female      0    81
2 female      1   233
3 male        0   468
4 male        1   109
```

```
Console Terminal Background Jobs
R 4.4.1 - /#
> titanic %>% group_by(Pclass, Survived) %>% summarise()
# A tibble: 6 x 3
# Groups:   Pclass [3]
  Pclass Survived n()
<dbl>    <dbl> <int>
1     1         0    80
2     1         1   136
3     2         0    97
4     2         1    87
5     3         0   372
6     3         1   119
```

```
Console Terminal Background Jobs
R 4.4.1 - /#
> titanic %>% group_by(Embarked, Survived) %>% summarise()
# A tibble: 7 x 3
# Groups:   Embarked [1]
  Embarked Survived n()
<chr>    <dbl> <int>
1 C         0    75
2 C         1    93
3 Q         0    47
4 Q         1    30
5 S         0   427
6 S         1   217
7 NA        1     2
```

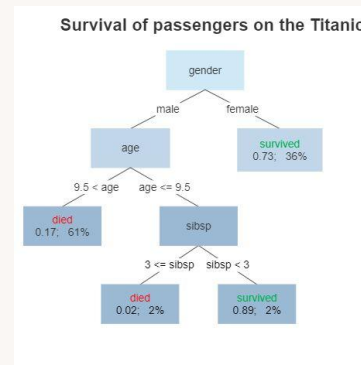
```
Source: lazy data [??] x 3
  sex_one_rule = ifelse(titanic$Sex == "female", 1, 0)
  emb_one_rule = ifelse(titanic$Embarked == "C", 1, 0)
  cls_one_rule = ifelse(titanic$Pclass == 1, 1, 0)
  non(titanic$Survived == sex_one_rule, na.rm = TRUE)
[1] 0.7867565
  non(titanic$Survived == emb_one_rule, na.rm = TRUE)
[1] 0.6377953
  non(titanic$Survived == cls_one_rule, na.rm = TRUE)
[1] 0.4798323
```

Right now, we're only using one variable to predict an outcome. How do we account for other variables in making our prediction? We can use a simple decision tree.

## Decision trees



A decision tree is a non-parametric supervised learning algorithm, which is utilized for both classification and regression tasks. It has a hierarchical, tree structure, which consists of a root node, branches, internal nodes and leaf nodes.



## Building blocks of decision trees >>

Entropy: measures how much information is present in a node.

$$\text{Entropy}(t) = - \sum_{i=1}^n p_i \log_2 p_i$$

Information gain: amount of information gained when a node is split into children nodes.

$$\text{Information gain}(t) = \text{Entropy}(t) - \sum_{i=1}^n \frac{n_i}{n} \text{Entropy}(t_i)$$

Supervised learning – classification

## Building blocks of decision trees >>

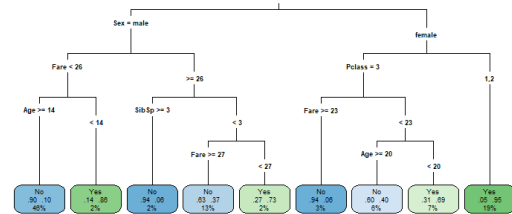
- Calculate entropy of the target variable
- Calculate entropy of the predictors
- Compute information gain by subtracting the weighted average entropy of predictor from entropy of the response
- Select the feature with the highest information gain and perform split
- Repeat recursively

Supervised learning – classification

## Decision trees on titanic



Sample decision tree model  
from Titanic dataset.



September 11, 2026

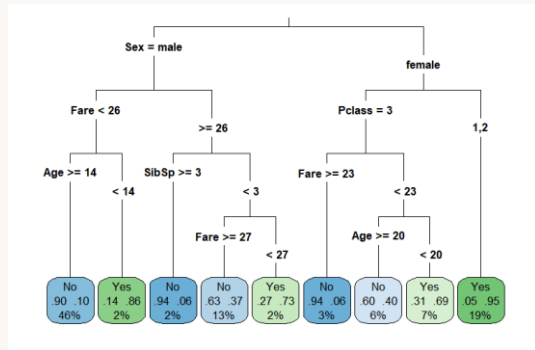
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## Supervised learning – classification

# Interpretation of decision tree >>

```
Console Terminal Background Jobs
R 4.4.1 ->
> tree_model
n= 712
node), split, n, loss, yval, (yprob)
= denotes terminal node
1) root 712 273 No (0.61657303 0.38342697)
2) sex=male 467 92 No (0.80299786 0.19700214)
4) Fare<= 26.28875 341 45 No (0.88803519 0.11964481)
8) Age>=14 327 33 No (0.89908257 0.10091743) *
9) Age<= 14 24 2 Yes (0.14285714 0.85714286) *
5) Fare>=26.28875 126 47 No (0.62698413 0.37301587)
10) SibSp>=2.5 17 1 No (0.94117647 0.05882353) *
13) SibSp<= 2.5 109 46 No (0.57798165 0.42201835)
22) Fare>=26.775 94 35 No (0.62765957 0.37234043) *
23) Fare<= 26.775 15 4 Yes (0.26666667 0.73333333) *
3) sex=female 245 64 Yes (0.26122449 0.73877551)
6) Pclass<= 110 53 No (0.51813187 0.48186813)
12) Fare>=23.35 18 1 No (0.94444444 0.05555556) *
13) Fare<= 23.35 92 40 Yes (0.43478261 0.56521739)
26) Age>=19.5 40 16 No (0.60000000 0.40000000) *
27) Age<= 19.5 52 16 Yes (0.30769231 0.69230769) *
7) Pclass<=1,2 135 7 Yes (0.05185185 0.94814815) *
> rpart.plot(tree_model)
>
```



September 11, 2026

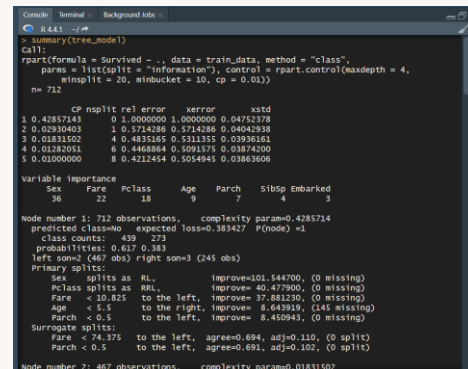
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# Interpretation of decision tree >>

**summary** function provides more details from a model object.

- Variable importance
- Complexity parameters
- Improvements on each node



```
summary(tree_model)
Call:
rpart(formula = Survived ~ ., data = train_data, method = "class",
      parms = list(split = "information"), control = rpart.control(maxdepth = 4,
      minsplit = 20, minbucket = 10, cp = 0.01),
      n = 712)

      CP nsplit rel error   error   xstd
1 0.42857143  0 1.0000000 1.0000000 0.04712378
2 0.02930403  1 0.5714286 0.5714286 0.04042938
3 0.01811502  4 0.4835185 0.5311355 0.03961611
4 0.01327051  6 0.4468844 0.5091575 0.03874200
5 0.01000000  8 0.4212454 0.5054945 0.03863606

Variable importance
Sex      Fare   PClass  Age   Parch  SibSp  Embarked
36      22      18      9      7      4      3

Node number 1: 712 observations, complexity param=0.4285714
predicted classno. expected loss=0.381427 P(node)=1
class counts: 439 273
probabilities: 0.617 0.383
left son=2 (467 obs) right son=3 (245 obs)
Primary splits:
Sex splits as RL, improve=101.544700, (0 missing)
Pclass splits as RRL, improve=40.477900, (0 missing)
Fare < 10.825 to the left, improve=37.881230, (0 missing)
Age < 5.5 to the right, improve= 8.643919, (45 missing)
Parch < 0.5 to the left, improve= 8.450943, (0 missing)
Surrogate splits:
Fare < 74.375 to the left, agree=0.694, adj=0.110, (0 split)
Parch < 0.5 to the left, agree=0.691, adj=0.102, (0 split)
Node number 2: 467 observations, complexity param=0.01811502
```

## Classification metrics



- Accuracy: the ratio of correct guesses to total guesses; works best when classes are balanced.
- Precision: the share of positive guesses that were actually correct; use when false positives are costly.
- Recall: the share of actual positives the model managed to find; use when missing a positive case is dangerous.
- F1 Score: the balanced middle ground (harmonic mean) between precision and recall.

## Classification metrics



- True positive (TP), true negative (TN), false positive (FP), false negative (FN)

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

## Imbalanced dataset



- When you have unequal distribution (99:1, 90:10) of the target classes, you have an **imbalanced dataset**.
- Some ways to address this:
  - Stratified sampling: divides a dataset into subgroups (strata) based on class labels
  - Undersampling: randomly drop majority rows to balance the ratio
  - Oversampling: duplicates existing minority instances randomly; fast, but can cause overfitting.

## Multi-label classification



- When you have more than 2 target classes, you are working on a multi-label classification problem.
- Examples:
  - Iris flower dataset
  - Classifying different fruits or animals
  - Object detection
- Classification metrics are still the same, but we need to do either macro or micro averaging.

# Regression problem

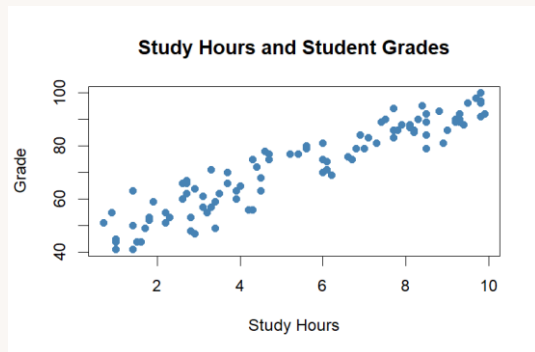
Learn how we analyze data, and build machine learning models for predicting continuous response

Supervised learning – regression

## How to estimate numeric values? >>

Here's a dataset showing hours rendered on studying and their respective grades.

"How much hours should I study if I want to get x grade?"

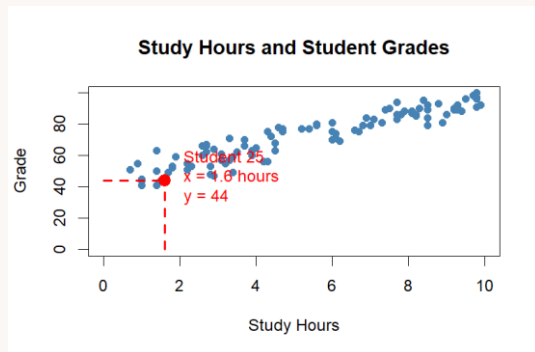


Supervised learning – regression

## How to estimate numeric values? >>

From the existing dataset, we can just “trace” the grades that we can get.

But not all students perform the same.

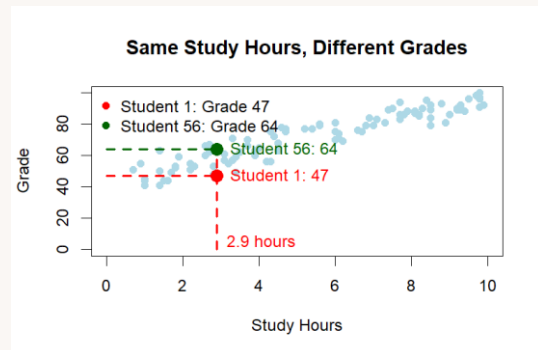


## Supervised learning – regression

# How to estimate numeric values? >>

Some students studied for the same number of hours, but got different scores.

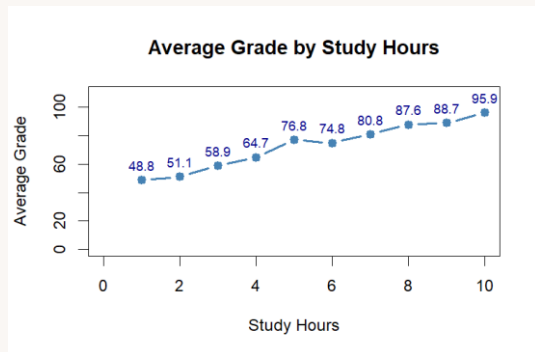
How can we estimate for cases like this?



## Supervised learning – regression

# How to estimate numeric values? >>

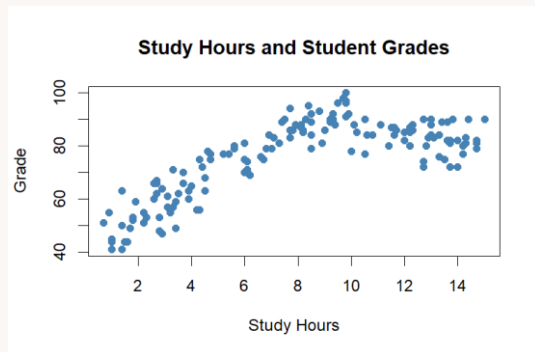
Here we can see the average grade based on the hours rendered on studying rounded to the nearest whole number.



Supervised learning – regression

## How to estimate numeric values? >>

What if we studied beyond 10 hours? How would our grade look like?

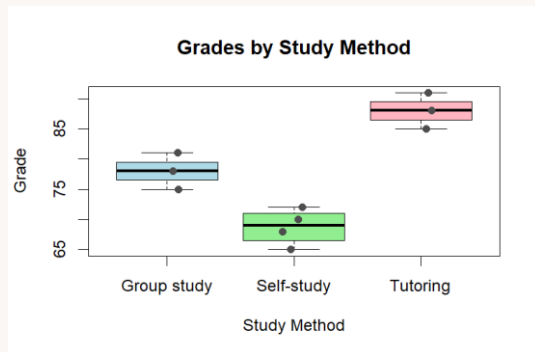


Supervised learning – regression

## How to estimate numeric values? >>

Another illustration of estimating our grades based on a categorical variable, e.g. study method.

If I get a tutor, would I get a higher grade?



## Introduction to linear regression >>

Linear regression models the relationship between one or more independent variables and a single dependent variable.

$$\hat{Y} = \beta_0 + \beta_1 X$$

Where, X is the independent variable, Y is the dependent variable,  $\beta_0$  is the intercept,  $\beta_1$  is the slope. We get the  $\beta_0$  and  $\beta_1$  values where:

$$\text{Minimize } \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Supervised learning – regression

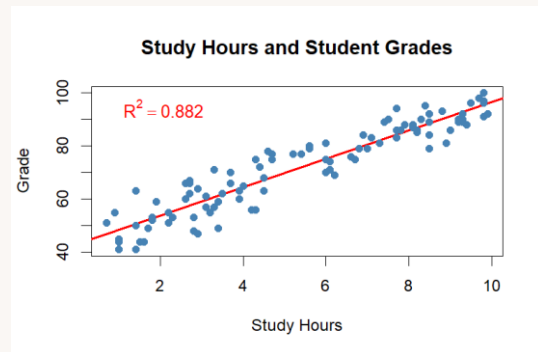
## Introduction to linear regression >>

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}, \beta_1 = r \left( \frac{\sigma_y}{\sigma_x} \right)$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

## Introduction to linear regression >>

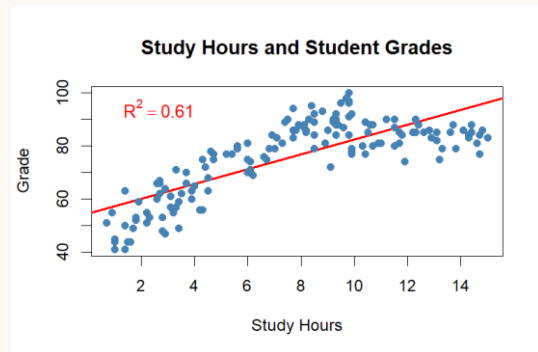
Here's the regression line fitted on the data. Generally, we see that many points are close to the line. For regression models, we want the points to be close to the regression line.



Supervised learning – regression

# Introduction to linear regression >>

What if we include the data points that aren't linear?



## Interpreting regression params >>

- Coefficients: (Intercept, X)
- Significance
- Multiple R-squared
- Adjusted R-squared
- p-value

```
Console Terminal Background Jobs
R 4.4.1 ~/
> summary(model)

Call:
lm(formula = grades ~ study_hours)

Residuals:
    Min       1Q   Median       3Q      Max
-12.3648  -3.9646  -0.1006   4.1261  12.3088

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  43.2197     1.2146   35.58  <2e-16 ***
study_hours   5.3368     0.1974   27.04  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.587 on 98 degrees of freedom
Multiple R-squared:  0.8818,    Adjusted R-squared:  0.8806
F-statistic: 731 on 1 and 98 DF,  p-value: < 2.2e-16

> |
```

## Regression metrics



- R-squared ( $R^2$ ): Measures the proportion of variance explained by the model, typically 0–1.
- MAE (Mean Absolute Error): Measures average absolute distance; linear, unit-consistent, and robust to outliers.
- MSE (Mean Squared Error): Averages squared errors, penalizing outliers heavily. RMSE (Root Mean Squared Error): Square root of MSE; penalizes large errors while remaining in the target unit.
- MAPE (Mean Absolute Percentage Error): Expresses error as a percentage, useful for relative comparisons across scales.

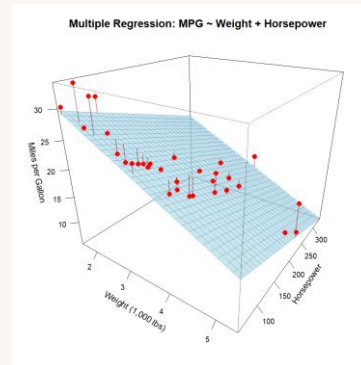
Supervised learning – regression

## Multiple regression model



We can extend linear regression to use multiple variables.

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_m X_m$$



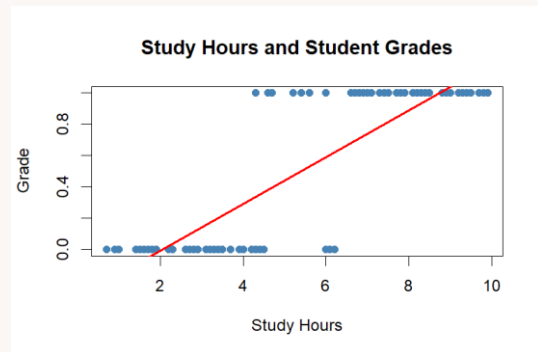
Supervised learning – regression

## Regression for classification?

>>

We can convert the student grades to pass/fail and make it binary classification problem and predict probabilities.

Ordinary linear regression model is problematic because the predictions can be negative or beyond 1.

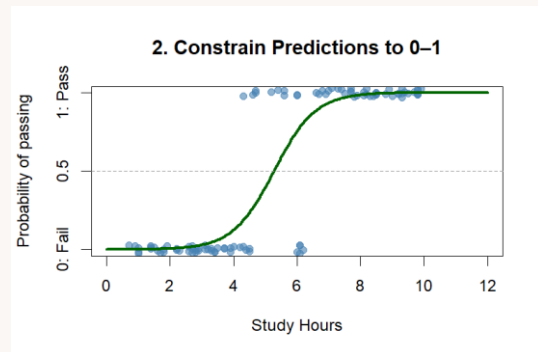


Supervised learning – regression

## Regression for classification?

>>

We need to force the values to be within the 0-1 range using the sigmoid function.



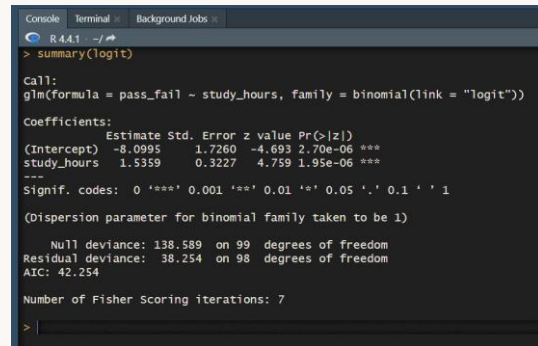
## Introduction to logistic regression >>

**Logistic regression** is a supervised machine learning algorithm used to predict the probability of a binary outcome (such as yes/no, 0/1, or true/false). Instead of fitting a straight line like linear regression, it uses an S-shaped sigmoid function to map input data to a value between 0 and 1.

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X \text{ or } P(Y = 1|X) = \frac{1}{1-e^{-(\beta_0+\beta_1 X)}}$$

## Interpreting logistic regression >>

- Log-odds >> odds >> probability
- Coefficients: (Intercept, X)
- Significance



```
Console Terminal Background Jobs
R 4.4.1 ~ /
> summary(logit)

Call:
glm(formula = pass_fail ~ study_hours, family = binomial(link = "logit"))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -8.0995     1.7260  -4.693 2.70e-06 ***
study_hours    1.5359     0.3227   4.759 1.95e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 138.589  on 99  degrees of freedom
Residual deviance:  38.254  on 98  degrees of freedom
AIC: 42.254

Number of Fisher Scoring iterations: 7

>
```

# Unsupervised Learning

Use machine learning models analyzing unlabeled data

Unsupervised learning

# Unsupervised learning



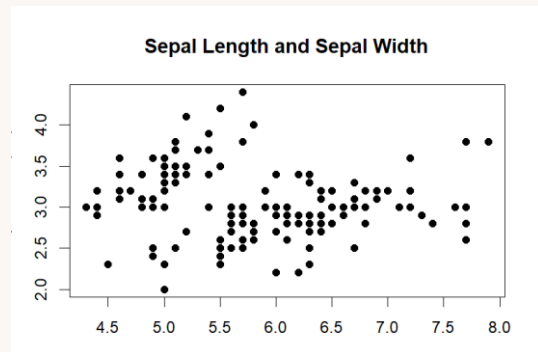
Unsupervised learning is a class of predictive analytics where the prediction generated is not based on a response variable. These are mostly used for clustering, dimensionality reduction, and association.

Unsupervised learning – clustering

# Cluster analysis



Cluster analysis attempts to group similar records based on their attributes.

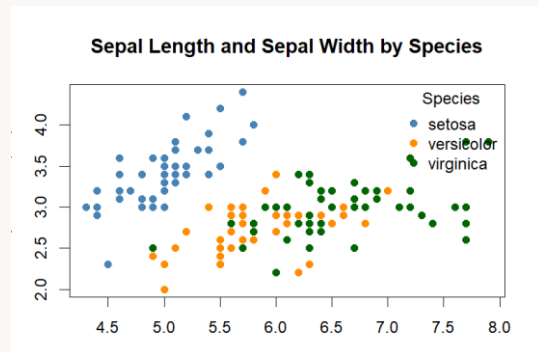


Unsupervised learning – clustering

# Cluster analysis



Cluster analysis attempts to group similar records based on their attributes.

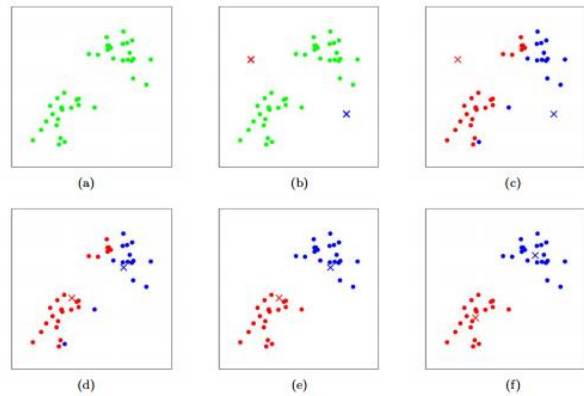


## Unsupervised learning – clustering

# K-means clustering



K-means clustering is an unsupervised machine learning algorithm that groups unlabeled data into k distinct clusters based on feature similarity by iteratively calculating center points called centroids.



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### Stopping Criteria for K-means

The K-Means algorithm does **not run forever**. It stops when **any one** of the following conditions is met:

#### ✅ 1. Centroids Do Not Change

If the centroids remain the same even after multiple iterations, It means the algorithm has **converged**.

#### ✅ 2. Points Remain in the Same Cluster

If no data point changes its cluster membership, The clusters are considered **stable**.

#### ✅ 3. Maximum Number of Iterations Is Reached

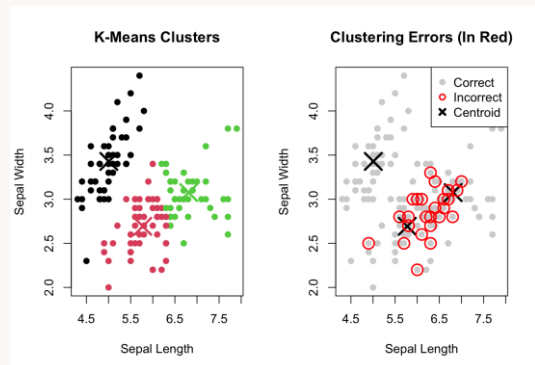
A maximum limit (e.g., 100 iterations) is set in advance. Once this limit is reached, the algorithm stops automatically.

Unsupervised learning – clustering

# K-means clustering



K-means clustering on iris dataset using sepal length and width.



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Unsupervised learning – clustering

# K-means clustering



K-means clustering on iris dataset using petal length and width.



Unsupervised learning – dimensionality reduction

## Dimensionality reduction



Dimensionality reduction is the process of transforming data from a high-number space into a lower-number space while keeping the most important details.

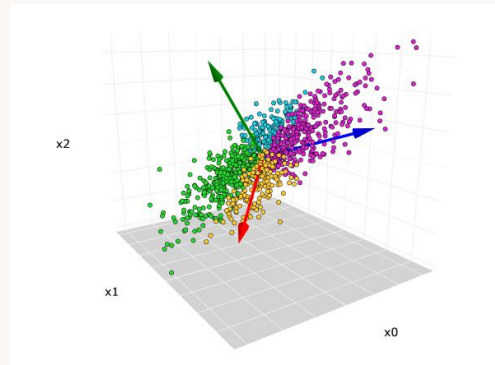
- Reduce redundancy
- Make the data easier to visualize
- Simplify downstream models
- Reduce noise

Unsupervised learning – dimensionality reduction

# Principal component analysis

>>

- Reduces the number of variables in a large dataset.
- Keeps the most important information and throws away noise.
- Creates new summary variables called principal components.



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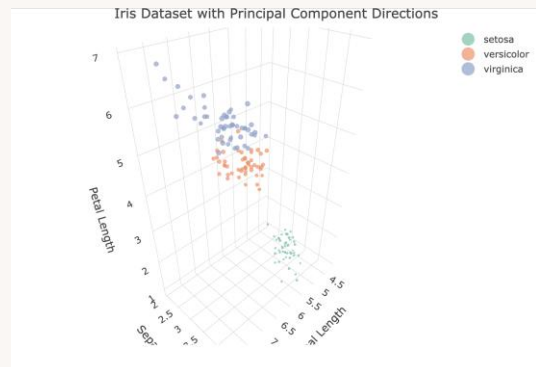
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Unsupervised learning – dimensionality reduction

## PCA on iris dataset



- The iris dataset has 4 variables: sepal length, sepal width, petal length, petal width



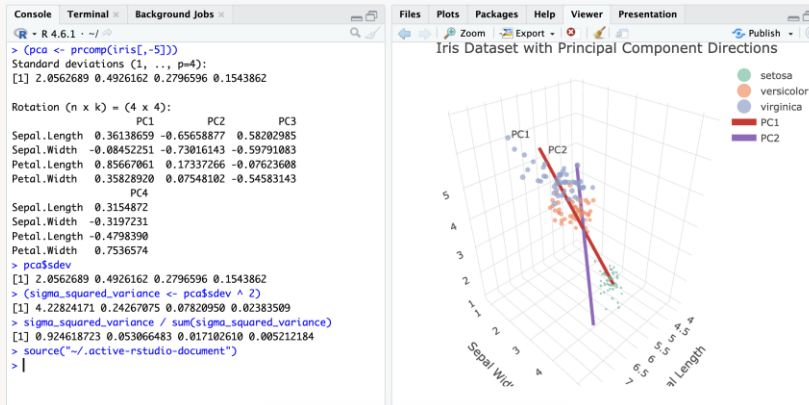
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## Unsupervised learning – dimensionality reduction

# PCA on iris dataset



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## Unsupervised learning – dimensionality reduction

# PCA on iris dataset

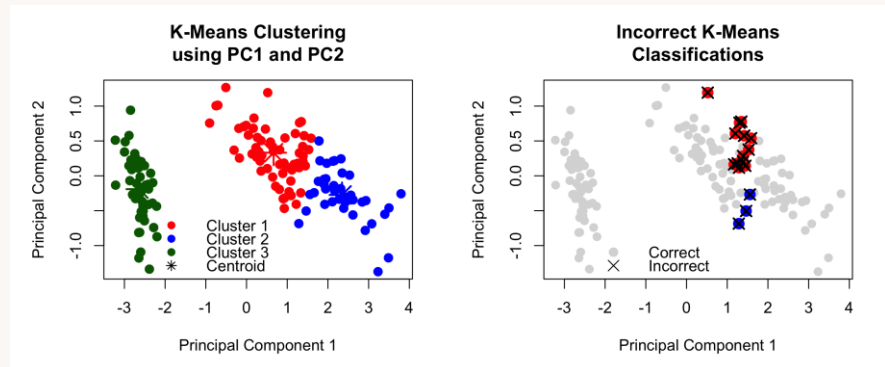


Accuracy:

Km1: 0.82

Km2: 0.96

Km3: 0.89



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# Prescriptive analytics

Turning business problems into mathematical models  
and leveraging data for better decisions

Prescriptive analytics

## Prescriptive analytics



**Prescriptive analytics** is the highest form of analytics. It answers the questions “what is the best decision?”, “what should we do?”, “if I do this, what will most likely happen?”.

Prescriptive analytics uses data, predictions, objectives, and constraints to **recommend actions**.

## Prescriptive analytics



"Because of the incoming Christmas season, demand for puto bumbong and bibingka will increase by 30% during ber-months."

- How much additional ingredients should we buy starting September?
- How many servings should we prepare each day?
- Do we need additional staff?
- How much budget should we allocate?
- What if actual demand is lower than predicted?

## Scientific decision-making >>

**Decision science** is the systematic study of how individuals and organizations make better choices.

- It focuses on making optimal choices based on available information.
- It values the judgments underlying these decisions, and to identify tradeoffs that might accompany any particular action or inaction.

## Components of a decision problem>>

- Objective: goals or desired results that we want to achieve (minimize or maximize)
- Alternatives: list of options that we have; advantages / disadvantages (payoffs)
- Constraints: limitations when making decisions

## Decision environments



- **Decisions under certainty:** occurs when the decision maker knows the outcome of each alternative; no uncertainty about future conditions.
- **Decisions under risk:** occurs when outcomes are uncertain, but the decision maker can assign probabilities to future events.
- **Decisions under uncertainty:** occurs when outcomes are uncertain and probabilities cannot be reliably assigned.

## Decisions under certainty



Example: A bibingka and puto bumbong stall needs to choose a supplier for the Christmas season. The prices and delivery times are **known**.

| Alternative | Ingredient Cost | Delivery Time |
|-------------|-----------------|---------------|
| Supplier A  | ₱5,000          | 2 days        |
| Supplier B  | ₱6,000          | 1 day         |

- If the objective is to **minimize cost** → choose **Supplier A**
- If the objective is to **minimize delivery time** → choose **Supplier B**

What if you need the supplies tomorrow, but don't have enough capital?

## Decisions under risk



Example: A food stall needs to decide how much bibingka and puto bumbong to prepare for next week. Based on previous Christmas seasons:

- If customer traffic is normal ( $p=40\%$ ), expected sales are 100 orders.
- If customer traffic is high ( $p=60\%$ ), expected sales are 200 orders.
- Preparing too much food may result in unsold inventory and wasted ingredients.

The owner therefore wants to estimate demand before deciding how much to prepare.

# Decisions under risk



Expected demand:

$$\sum_{i=1}^n x_{ij} p_j$$

- $x_{ij}$ : payoff for decision  $i$  in state  $j$
- $p_j$ : probability of  $j$  happening

## Decisions under risk



Expected demand:

$$\text{Expected demand} = (0.4 \times 100) + (0.6 \times 200)$$

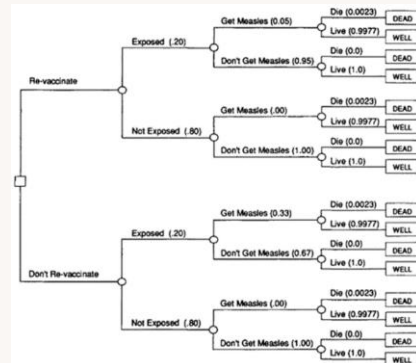
$$\text{Expected demand} = 40 + 120$$

$$\text{Expected demand} = 160$$

# Decisions trees



A decision tree is a map of the possible outcomes of a series of related choices. It allows an individual or organization to weigh possible actions against one another based on their costs, probabilities, and benefits.



## Decisions trees



Example: A small food business selling **bibingka and puto bumbong** is considering opening a temporary Christmas stall near a church during Simbang Gabi. The owner must first decide:

- **Do not open the stall**
- **Run a 3-day trial stall, costing ₱5,000**

Based on previous events, management estimates that the trial stall has:

- **70% probability of strong customer demand**
- **30% probability of weak customer demand**

## Decisions trees



If the trial is weak, the owner stops. If the trial is successful, the owner must decide whether to operate a:

- **Small Stall**, additional setup cost = ₱10,000
- **Large Stall**, additional setup cost = ₱20,000

## Decisions trees



After opening, actual Christmas demand can still be:

| Demand        | Probability |
|---------------|-------------|
| High Demand   | 60%         |
| Normal Demand | 40%         |

Estimated **net sales before setup costs** are:

| Demand Outcome | Small Stall | Large Stall |
|----------------|-------------|-------------|
| Normal Demand  | ₱25,000     | ₱30,000     |
| High Demand    | ₱40,000     | ₱60,000     |

Prescriptive analytics

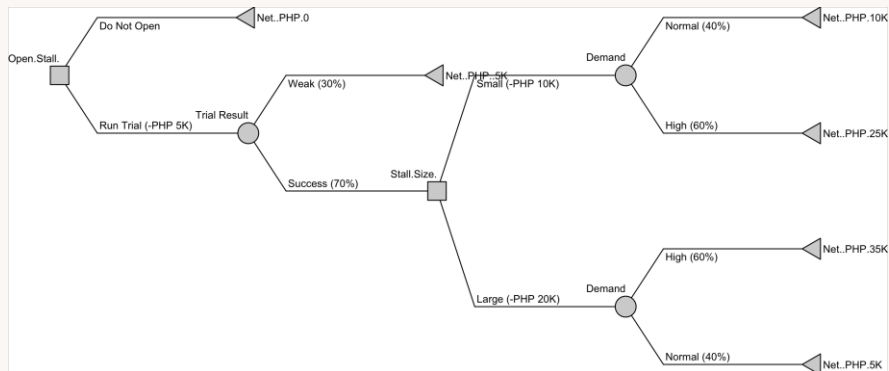
## Decisions trees



So the question becomes:

- **Should the owner conduct the trial, and if the trial is successful, should they open a small or large stall?**

# Decisions trees



## Expected monetary value



Acquiring the expected monetary value from each stall size.

$$EMV_{\text{Small}} = (0.40 \times \text{P}25,000) + (0.60 \times \text{P}40,000) - \text{P}10,000 = \text{P}24,000$$

$$EMV_{\text{Large}} = (0.40 \times \text{P}30,000) + (0.60 \times \text{P}60,000) - \text{P}20,000 = \text{P}28,000$$

If the trial is successful, shop owner should open a **large stall**.

## Expected monetary value



Acquiring the expected monetary value from the trial run.

$$\begin{aligned} EMV_{\text{Do Not Open}} &= \text{P}0 \\ EMV_{\text{Run Trial (Large)}} &= 0.70(\text{P}28,000 - \text{P}5,000) + 0.30(\text{P}0 - \text{P}5,000) \\ &= \text{P}14,600 \\ EMV_{\text{Run Trial (Small)}} &= 0.70(\text{P}24,000 - \text{P}5,000) + 0.30(\text{P}0 - \text{P}5,000) \\ &= \text{P}11,800 \end{aligned}$$

**The owner should conduct the trial.**

## Decisions under uncertainty >>

Example: A bibingka and puto bumbong business is deciding how aggressively to operate during the Christmas season. Management has three alternatives:

- **Aggressive Expansion** — larger stall, more inventory, more staff
- **Moderate Expansion** — additional inventory and limited additional staff
- **Conservative Operation** — maintain a small operation

## Decisions under uncertainty >>

However, management does **not know how strong customer demand will be**, and there is not enough reliable historical data to assign probabilities. Possible outcomes are:

- Weak Demand
- Moderate Demand
- Strong Demand

## Decisions under uncertainty



Estimated net profits, in ₱000s, are:

| Decision Alternative   | Weak Demand | Moderate Demand | Strong Demand |
|------------------------|-------------|-----------------|---------------|
| Aggressive Expansion   | -400        | 400             | 1,200         |
| Moderate Expansion     | -100        | 350             | 750           |
| Conservative Operation | 100         | 250             | 450           |

Prescriptive analytics

## Decisions under uncertainty



Maximax: the optimistic decision

- Get the maximum payoff per alternative
- Get the alternative with best maximum payoffs

Aggressive expansion (strong demand) – 1,200

Prescriptive analytics

## Decisions under uncertainty >>

Maximin: the conservative decision

- Get the minimum payoffs per alternative
- Get the alternative with the highest minimum payoff

Conservative operation (weak demand) – 100

## Decisions under uncertainty >>

Minimax regret: if I make the wrong choice, how much will I regret it?

- First determine the **best payoff for each demand condition**.
  - Weak: 100, moderate: 400, strong: 1,200
- Calculate regret = best payoff for state – actual payoff

| Decision Alternative   | Weak | Moderate | Strong | Maximum Regret |
|------------------------|------|----------|--------|----------------|
| Aggressive Expansion   | 500  | 0        | 0      | 500            |
| Moderate Expansion     | 200  | 50       | 450    | 450            |
| Conservative Operation | 0    | 150      | 750    | 750            |

Prescriptive analytics

## Decisions under uncertainty >>

- Then choose the decision with the **smallest maximum regret**.

Moderate expansion – 450

# References

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